Liancheng Gong (Krystal)

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Education

2025– **University of Maryland**, College Park, MD
Doctor of Philosophy in Information Studies, GPA: 4.00/4.00
Advised by Assistant Professor Julia Mendelsohn

2022-2024 **University of Pennsylvania**, Philadelphia, PA
Master of Science in Engineering in Data Science, GPA: 3.90/4.00

- **Math**: *Big Data Analytics*, *Statistics for Data Science*
- **Computer Science**: *Database and Information Systems*, *Intro to Software Development*
- **Artificial Intelligence**: *Principles of Deep Learning*, *Artificial Intelligence*, *Ethical Algorithm Design*

2018-2022 **New York University**, New York City, NY | Abu Dhabi, UAE | Shanghai, China
Bachelor of Science in Data Science, GPA: 3.81/4.00

- **Math**: *Probability and Statistics*, *Ordinary Differential Equations*, *Introduction to Math Modeling*
- **Core Data Science**: *Data Structures*, *Machine Learning*, *Information Visualization*, *NLP*, *Algorithms*
- **Human-Centered**: *Decision Making Under Uncertainty*, *Human-Centered Data Science*

Honors and Awards

2025–2026 Dean's Fellowship, University of Maryland

2022 Magna Cum Laude, New York University

2019–2022 Dean's List for Academic Year, New York University

Publications

2026 Abhinav Rao, **Liancheng Gong**, Bin Hu, Atharva Naik. *An Emergent Mirage: Is Emergent Misalignment and Realignment Indeed a Robust Phenomenon?* Preprint, under review. [arXiv:2607.09053]

2026 **Liancheng Gong**, Zhiyang Wang, Yiwei Xu, Julia Mendelsohn. *Persuasion Index: A Theory-Guided Framework for Persuasion Analysis.* Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP), Main Conference. [arXiv:2606.14580]

2026 **Liancheng Gong**, Wang Bill Zhu, Jesse Thomason, Li Zhang. *Iterative Formalization and Planning in Partially Observable Environments.* Findings of the Association for Computational Linguistics (ACL).

2026 **Liancheng Gong**, Daniel J. Hopkins, Samuel Wolken. *The Limits of De-politicizing—and Also of Annotation: A Case Study in Russian Newspaper Posts on Social Media 2016–2024.* Proceedings of the International AAAI Conference on Web and Social Media (ICWSM).

Conference Presentations

2026 "Persuasion Index: A Theory-Guided Framework for Persuasion Analysis." **Liancheng Gong**, Zhiyang Wang, Yiwei Xu, Julia Mendelsohn.

- Mid-Atlantic Student Colloquium on AI, Language, and Learning (MASC-ALL). Washington, D.C. April 25.
- International Conference on Computational Social Science (IC2S2). Burlington, VT. July 28–31.
- Text as Data (TADA). Berkeley, CA. Oct 5–6.

2026 "GeopoliScope: Tracing Geopolitical Stance Formation in Language Model Hidden Representations." Ashwath David, **Liancheng Gong**, Julia Mendelsohn.

- Mid-Atlantic Student Colloquium on AI, Language, and Learning (MASC-ALL). Washington, D.C. April 25.
- International Conference on Computational Social Science (IC2S2). Burlington, VT. July 28–31.

2026 "Iterative Formalization and Planning in Partially Observable Environments." **Liancheng Gong**, Wang Bill Zhu, Jesse Thomason, Li Zhang. Association for Computational Linguistics (ACL). San Diego, CA. July 2–7.

2026 "The Limits of De-politicizing—and Also of Annotation: A Case Study in Russian Newspaper Posts on Social Media 2016–2024." **Liancheng Gong**, Daniel J. Hopkins, Samuel Wolken. International AAAI Conference on Web and Social Media (ICWSM). Los Angeles, CA. May 27–29.

Reviewing

2026 IC2S2, ACM TIST, AAAI

2025 AAAI

Funding

2026 **CRA-WP Grad Cohort Workshop**
Funding source: CRA Scholarship

Research Experience

2024–2025 **Drexel University**, Department of Computer Science, Philadelphia, PA
Researcher, working with Assistant Professor Harry Zhang

- Structured **Large Language Model (LLM)** outputs into **Planning Domain Definition Language (PDDL)** domain and problem files to generate verifiable plans.
- Iteratively refined outputs using environmental feedback to improve generation accuracy.
- Categorized recurring errors across trials, experimenting with solutions including memory integration and partial file editing to enhance model robustness.

2024–2025 **University of Pennsylvania**, Department of Political Science, Philadelphia, PA
Research Coordinator (full-time), working with Professor Daniel Hopkins

- Analyzed polls and exit polls for FiveThirtyEight's 2024 presidential election coverage.
- Investigated Democratic and Republican advertisement campaigns across Meta, Instagram, and Google; examined topic trends in Russian media outlets, such as anti-government narratives.
- Scraped and processed 2M+ VKontakte posts from 19 media outlets via **HPC**, created feature vectors using **TF-IDF and BERT** embeddings, and developed a **logistic regression model** to identify political posts, increasing the sample's political content from 10% to 50%.
- Annotated Russian blurbs via **OpenAI API**, including prompt engineering for annotation and translation, achieving 90% alignment between GPT and human annotations on political features.
- Classified symbolic rhetoric by generating embeddings and fine-tuning on **BERTweet** to detect topics like "us vs. them" and abstract themes.

2024 **University of Pennsylvania**, Computational Social Science Lab, Philadelphia, PA
Data Scientist, working with Yuxuan Zhang and Regina (Jeanne) Ruane

- Designed a master list to annotate and analyze bias in media discussions on topics like Biden's age and NATO, delivering findings to the dashboard.
- Automated **extract, transform, and load (ETL)** processes for streaming Nielsen data, improving data consistency and minimizing manual handling.
- Managed an **AWS-hosted database**, ensuring data accuracy and schema consistency; standardized network names across large datasets for enhanced data cohesion and analysis.

2023 **University of Pennsylvania**, Computational Social Science Lab, Philadelphia, PA
Graduate Research Assistant

Atlas, working with Mark Whiting

- Segmented and mined information from PDFs, leveraging **OpenAI's question-answering API** and context-window techniques, e.g., Dynamic Memory Networks and Parallel Context Windows.
- Pioneered an **LLM** approach by refining prompts and GPT models, achieving superior comprehension and context alignment.
- Sourced a robust dataset of 300+ papers and crafted 300+ precise questions, covering all critical analysis units and elevating the accuracy and depth of the literature analysis.

Human Mobility Data for Epidemiology, working with Jorge Barreras Cortes

- Developed **Susceptible-Exposed-Infected-Recovered models** on contact networks utilizing **Poisson Distribution** and **Tensor** methods to predict virus spread, guiding intervention strategies.
- Managed **ETL** of 100GB+ databases via **AWS RDS and EMR**; conducted cross-analysis on POI distribution and mobility indicators, ensuring model applicability and accuracy.
- Optimized the **DBSCAN** clustering algorithm on population data, providing insights into area-specific visit patterns.

Projects

Machine Q&A on Chinese Judicial Reading Comprehension [Code] [PDF]
Developed a Span-Extraction Machine Reading Comprehension (MRC) system.

Colorize Images with Deep Learning [Code] [PDF]
Compared ResNet, GAN, and VAE approaches to colorize greyscale images.

Customer Lifecycle, NYU X HSBC [Code] [PDF]
Segmented customers, predicted inactivity, and provided targeted recommendations for HSBC.

A Model to Predict the Maximum Car Speed in the Rain [Code] [PDF]
Quantified driver visibility and calculated the maximum safe speed under rainy conditions.

Teaching Experience

2022–2023 **Fife Academy Coding Club**, University of Pennsylvania, Philadelphia, PA
Coding Instructor

- Developed and led a modular coding curriculum for 100+ K–12 students, with a 100% positive feedback rate, delivering weekly lectures, hands-on labs, and individual feedback.

Industry Experience

2021 **Tencent Technology (Shenzhen) Co., Ltd**, Shanghai, China
Data Scientist, Platform and Content Group

- Enhanced user segmentation and ad CTR by 5% through analysis of 100M+ user records (Python, MySQL); developed ARIMA (0.073 RMSE) for activity forecasting and used NLP on search logs, increasing retention by 4% via A/B testing.

2020–2021 **Shanghai Urban Real Estate Appraisal Co., Ltd**, Shanghai, China
Data Analyst, Data R&D Department

- Developed predictive models (Regression, Random Forest, XGBoost) with 80% R^2 for rental pricing, supported by literature review, a survey with 300+ responses, and a 2,000+ entry database from field visits and web scraping.

Computer Skills

Programming Languages: Python, R, SQL, Java

Frameworks and Libraries: NumPy, Pandas, SciPy, Scikit-Learn, PySpark, Seaborn, Matplotlib, PyTorch, TensorFlow

Tools and Platforms: AWS (RDS/SageMaker/S3/EMR), MySQL, MongoDB, Tableau, Power BI, Git, Linux, Stata, Matlab, ETL